

Body Composition Assessment Method Comparison



Method	Accuracy*	Pros	Cons
Bioelectrical impedance analysis (BIA), bioimpedance spectroscopy (BIS), and electrical impedance myography (EIM)	3–4, depending on model	• Simple and quick • Affordable (some models) • Portable (some models)	• Limited availability (some devices) • Variable accuracy
Skinfolds	3–4, depending on assessor	• Affordable • Fairly simple and quick • Portable • Provides regional information	• Less accurate • Unpleasant for some individuals
Ultrasound	3	• Fairly simple and quick • Provides regional information	Limited availability
Circumferences	4	• Affordable • Simple and quick • Portable • Provides limited regional information	Less accurate
Underwater weighing (hydrodensitometry)	2	Accuracy	• Limited availability • Expensive • Not portable • Unpleasant for some individuals
Air displacement plethysmography (ADP)	2	• Accuracy • Fairly simple	• Limited availability • Expensive • Not portable

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Dual-energy X-ray absorptiometry	1	• Accuracy • Quick • Provides regional information • Provides bone density	• Very-limited availability • Expensive • Not portable • Small amount of radiation
Infrared 3-dimensional scanning	N/A	• Simple and quick • Visualization of body shape	• Not portable • Limited information on accuracy
*Rating from 1 (most accurate) to 4 (least accurate)			